

Visual Search in Amblyopia: Abnormal Fixational Eye Movements and Suboptimal Sampling Strategies

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PURPOSE. Microsaccades shift the image on the fovea and counteract visual fading. They are also thought to serve as an optimal sampling strategy while viewing complex visual scenes. The goal of our study was to assess visual search in amblyopic children.

METHODS. Twenty-one amblyopic children with varying severity of amblyopia and 10 healthy controls were recruited. Eye movements were recorded using infrared video-oculography during amblyopic and fellow eye viewing while the subjects performed (1) visual fixation, (2) exploration of a blank scene, and (3) visual search task (spot the difference between two images). The number of correctly identified picture differences and reaction time were recorded. Microsaccade, saccades, and intersaccadic drifts were analyzed in patients without latent nystagmus (LN). Slow phase velocities were computed for patients with LN.

RESULTS. Both patients with and without LN were able to spot the same number of differences but took longer during fellow eye viewing compared to controls. The ability to identify differences was diminished during amblyopic eye viewing particularly those with LN and severe amblyopia. We found reduced frequencies of microsaccades and saccades in both amblyopic and fellow eyes during fixation and visual search but not during exploration of blank scene. Across all tasks, amblyopes with LN had increased intersaccadic drifts.

CONCLUSIONS. Our findings suggest that deficient microsaccade and saccadic activity contributes to poorer sampling strategy in amblyopia, which is seen in both amblyopic and fellow eye. These deficits are more notable among subjects who experienced binocular decorrelation earlier in life, with subsequent development of LN.

Keywords: amblyopia, visual search, microsaccades, saccades, nystagmus

Vision is a dynamic process that relies on the near constant motion of the eyes. During visual exploration, saccades alternate with foveal fixations in a goal-directed fashion.¹ Saccades shift the gaze to bring objects of interest within the high-resolution foveal region of the eye. Microsaccades, which are small rapid shifts, have been shown to keep objects within a retinal locus of fixation.² Based on studies in normal subjects and neural recordings in nonhuman primates, microsaccades are thought to play a role in counteracting visual fading³ and in visual search. Microsaccades provide an optimal sampling method for the brain when viewing fine objects, sharp edges, and contrast details.^{4–8}

Amblyopia occurs due to de-correlated binocular input to the visual cortex. Functional and anatomic abnormalities in cortical area V1 are well known in amblyopia.^{9–16} However, these V1 losses do not account for the full range of perceptual deficits.^{9,17,18} Subcortical eye movement sensitive areas controlled by V1 via extrastriate visual pathways are also affected.¹⁹ Amblyopes are known to have increased fixation instability²⁰ and difficulties initiating and executing saccades,²¹ primarily in the amblyopic eye. Furthermore, those who have experienced binocular decorrelation in early infancy, are known to develop

fusion maldevelopment nystagmus (FMN) or latent nystagmus (LN).^{22–24} In prior research, we reported reduced microsaccade frequency during visual fixation that was more prominent with increased amblyopic severity,²⁵ as well as subtle gaze holding deficits of the fellow eye in patients without LN.²⁵ The occurrence of microsaccades correlates with the firing of cells in the striate cortex.^{26–29} Thus, the study of microsaccades in amblyopia represents a unique opportunity to elucidate their role in abnormal visual processing states.

In the present study, we quantified microsaccades, saccades, and intersaccadic drifts in amblyopic children during visual fixation, viewing of a blank scene and visual search. We hypothesize that patients without LN will have greater difficulties with visual search during both fellow and amblyopic eye viewing conditions due to reduced microsaccade production. We also predict that patients who experienced disruption of binocularity in early infancy who developed LN will have more pronounced difficulties with visual exploration due to increased slow phase drift. Some of these data have appeared before in abstract form (Wang Y-Z, et al. *IOVS* 2017;58:ARVO E-Abstract 2336).





FIGURE 1. Example of visual search activity trial image. Picture differences are circled in yellow. Subjects were asked to identify and click on picture differences. In analysis, images were divided into interest areas by a grid of 50 squares (pixels $\times$ pixels). Time and location of clicks were recorded. Image from the app Picture Mania Deluxe by Raccoon Digital Technology Co Ltd.

METHODS

We recruited 31 subjects (controls = 10 and amblyopia = 21). We assessed eye movement traces for presence of LN ($n = 10$). Amblyopia was classified by severity of disease as mild ($n = 9$), moderate ($n = 8$), and severe ($n = 4$), and based on etiology as strabismic ($n = 4$), anisometropic ($n = 10$), and mixed mechanism ($n = 7$) per Pediatric Eye Disease Investigator Group (PEDIG) studies.⁵⁰ Only one anisometropic subject had LN. All but two subjects with mixed/strabismic amblyopia had LN. As such, we did not analyze the data by type of amblyopia due to significant overlap in cohorts. Stereopsis was assessed using the Titmus Fly Stereotest (Stereopsis-Present: detect the fly or better [$n = 13$] and Stereopsis-Absent: unable to detect the fly [$n = 8$]). Only two patients had fine stereopsis (better than 60 sec arc). Both had mild anisometropic amblyopia with no LN.

Eye Movement Recordings

A high-resolution video-based eye tracker (EyeLink 1000) was used as described in our previous work.^{31,32} The experiment protocols complied with the tenets of the Declaration of Helsinki and were approved by the Cleveland Clinic Institutional Review Board. Written informed consent was obtained from the parents or legal guardians on behalf of all the children. Each eye was calibrated under monocular viewing with the best-corrected vision before the beginning of every trial. The tasks were performed monocularly, during fellow eye viewing (FEV) and amblyopic eye viewing (AEV). Eye movements of both viewing eye and nonviewing eye were analyzed. The viewing distance was 55 cm. Subjects participated in three tasks for 45 seconds each: (1) visual fixation: subjects were instructed to fixate on a red circular target that subtended a 0.5° visual angle on a white background; (2) free viewing of a blank scene: subjects were required to perform a visual exploration of a blank scene, which was 50% gray; and (3) visual search: subjects were instructed to identify

differences (10 total) between two similar looking images and click on them immediately, once found (Fig. 1). Subjects were not informed of the total number of picture differences. Time and location of these clicks were recorded. All images were equalized for average luminance except for those in the blank scene, which were 50% gray.

All analyses were performed in Matlab and GraphPad Prism 7 (La Jolla, CA, USA).

Behavioral Parameters

We divided each image set of the visual search trial into a grid of 50 total squares (interest areas), 25 squares/image. We analyzed parameters including correctly identified picture differences, reaction time to first identified difference, and visual exploration strategy factors (percent of first interest areas explored that correlated with first click, time from first look in an interest area to first click, percent of fixations in interest areas containing picture differences, percent of trial dwell time spent in correct interest areas). Fixation durations in each interest areas, from each trial, were compiled and percentiles from each subject were calculated. ANOVA was used to compare these variables across controls and amblyopic patients. An unpaired t -test was used to compare reaction time in patients with and without stereopsis. χ^2 tests were used to compare percent of trials in which the first interest areas explored corresponded to the area of the first click, across controls and amblyopes.

Oculomotor Parameters in Subjects Without LN

Microsaccades and saccades were detected using the Engbert and Kliegl algorithm.³³ Microsaccades were defined as saccades with an amplitude of less than or equal to 1 degree.^{10,28,34–36} We used composite vectors of horizontal and vertical eye movements for analysis. We used paired t -tests to compare parameters obtained during FEV versus AEV during visual fixation and blank scene viewing. For the visual search,

TABLE 1. Clinical and Demographic Characteristics of the Study Population

Category	Sex	Age	Acuity OD	Acuity OS	Stereopsis	Spherical Equivalent OD	Spherical Equivalent OS	Strabismus Near	Strabismus Distance	LN
Control	F	13	0	0	40 s	−4.00	−4.00	ortho	ortho	None
Control	F	8	0	0	40 s	+4.50	+4.50	ortho	ortho	None
Control	F	10	0	0	40 s	+1.25	+1.25	ortho	ortho	None
Control	F	15	0	0	40 s	−3.25	−3.25	ortho	ortho	None
Control	F	11	0	0	40 s	−2.25	−2.50	ortho	ortho	None
Control	M	19	0	0	40 s	−5.50	−5.25	ortho	ortho	None
Control	M	8	0	0	40 s	−2.75	−3.00	ortho	ortho	None
Control	F	12	0	0	40 s	−3.25	−3.75	ortho	ortho	None
Control	M	12	0	0	40 s	−1.25	−1.25	ortho	ortho	None
Control	F	7	0	0	60 s	+1.00	+1.25	ortho	ortho	None
Mild aniso	F	8	−0.12	0.17	100 s	0.00	+3.50	ortho	ortho	None
Mild aniso	M	10	−0.12	0.17	60 s	−0.25	+4.50	ortho	ortho	None
Mild aniso	F	9	0	0.30	50 s	−1.25	−3.00	ortho	ortho	None
Mild aniso	M	9	0	0.17	400 s	+1.5	+1.25*	ortho	ortho	Yes
Mild mixed	F	5	0.17	0	3000 s	+4.00	+2.25	25 X(T)	30 X(T)	Yes
Mild mixed	M	10	0	0.17	400 s	+4.25	+5.25	10 E(T)	10 E(T)	Yes
Mild mixed	F	8	0.17	−0.12	100 s	+3.75	+1.25	ortho	ortho	None
Mild mixed	F	10	0.17	0	Absent	+3.25	+2.25	4 X(T)	6 X(T)	Yes
Mild strabismic	M	11	0	0.17	Absent	+3.50	+3.00	2-4 E(T)	2-3 E(T)	Yes
Mod aniso	M	6	0	0.48	200 s	+0.25	+5.25	ortho	ortho	None
Moderate aniso	M	16	0.39	0	400 s	−7.25	−2.25	ortho	ortho	None
Moderate aniso	F	9	0.48	0	200 s	+3.00	+0.25	ortho	ortho	None
Moderate aniso	F	8	0	0.60	Absent	+0.25	+5.00	ortho	ortho	None
Moderate mixed	M	15	−0.12	0.79	400 s	+0.50	+3.25	20-25 X(T)	20 -25X(T)	Yes
Moderate mixed	M	15	0.54	0	Absent	+4.50	0.00	ortho	ortho	Yes
Moderate strabismic	F	10	0	0.48	400 s	+2.50	+2.50	6 E(T)	Flick E(T)	None
Moderate strabismic	F	10	0.09	0.39	Absent	+2.75	+2.75	4 E(T)	4 E(T)	Yes
Severe aniso	F	12	0	1	Absent	+1	+6.75	ortho	ortho	None
Severe aniso	M	8	1	0	400 s	+6.5	0.00	ortho	ortho	None
Severe mixed	M	11	0.79	0	Absent	−8.25	+0.50	35-40 E(T)	35 E(T)	Yes
Severe strabismic	M	7	0.87	0	Absent	+2.25	+2.75	35 E(T)	40 E(T)	Yes

* The amblyopic eye had >+1.5 astigmatism than the fellow eye.

oculomotor parameters among amblyopes were compared to controls using ANOVA. To assess patterns of microsaccade production during varying levels of scene complexity, we analyzed microsaccade frequency across all three tasks using repeated-measures ANOVA.

Oculomotor Parameters in Subjects With LN

Quick phases of LN are known to share the same dynamic properties as microsaccades and volitional saccades.^{37,38} Thus, for data analysis, we pooled together the quick phases and saccadic events. We computed combined quick phases and saccadic frequencies, amplitudes, and intervals. We also calculated the composite slow phase velocity in patients with LN (periods between quick phases and saccades) and compared it to the intersaccadic drift velocity in patients without LN (periods between microsaccades and saccades). Twenty milliseconds of data were removed from the start and end of each period, to exclude acceleration and deceleration of the eye from fast eye movements. We analyzed the median slow phase velocity within and across each task and compared them to the intersaccadic drift of amblyopes without LN and controls, using a 1-way ANOVA. We used a Wilcoxon test to compare the median slow phase velocity during visual fixation and blank scene viewing. We used paired *t*-tests to compare the frequency and interval between the quick phases of LN for visual fixation and quick phases of LN/saccades for blank scene viewing. For visual search, we used a 1-way ANOVA to

compare the frequency of quick phases of LN/saccades, as a function of amblyopia severity. We used a paired *t*-test to compare the amplitudes of quick phases and saccades for each task.

For all tests in which an ANOVA was performed, Dunnett's multiple comparison post-hoc testing was also performed for significant results.

RESULTS

Table 1 lists the clinical features of the subjects. There was no difference between age ($P = 0.179$, *t*-test) and gender ($P = 0.24$, χ^2 test) between amblyopes and controls. We divided amblyopes into two cohorts based on presence or absence of LN and analyzed the data as a function of amblyopia severity. There was no difference between amblyopia severity ($P = 0.75$, χ^2 test) or stereopsis status in patients with and without LN ($P = 0.14$, χ^2 test).

Behavioral Performance

We analyzed the correctly identified picture differences during FEV and AEV. The number of correctly identified picture differences were similar during FEV irrespective of the presence of LN (Figs. 2A, 2B) or the severity of amblyopia (Figs. 3A, 3B). During AEV, LN patients identified fewer differences (control: 4.73 ± 2.5 ; LN absent: 2.82 ± 2.32 ; LN present: 2.50 ± 2.17 , $P = 0.0013$, ANOVA) and had greater

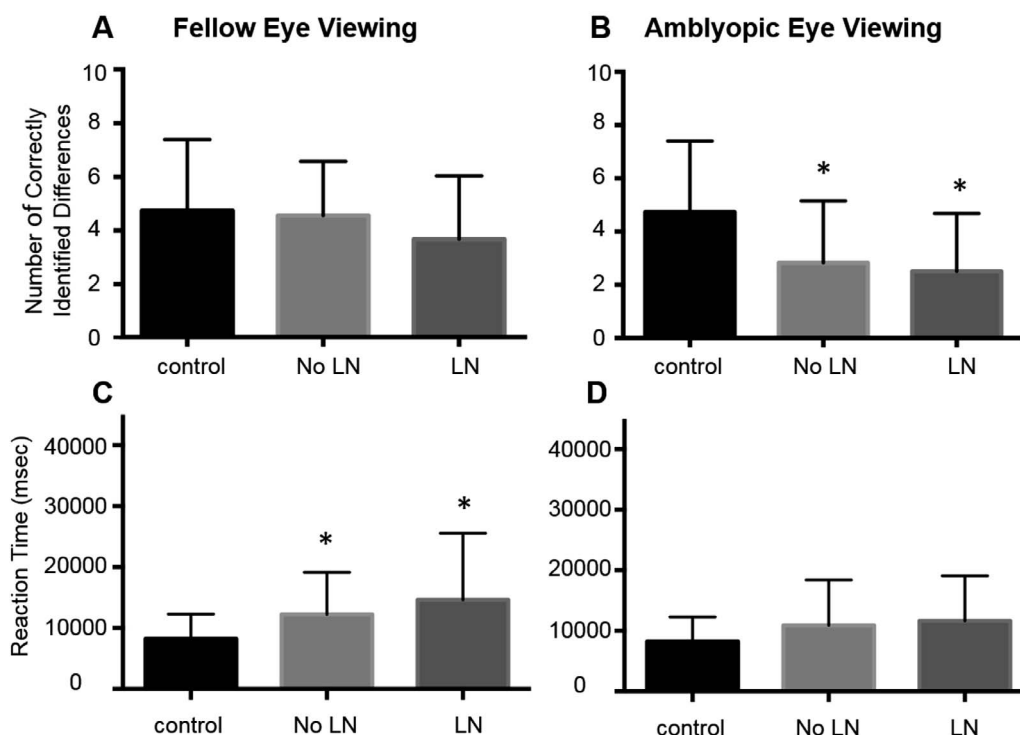


FIGURE 2. A and B represent correctly identified picture differences as a function of presence or absence of LN in the FEV condition and AEV condition, respectively. C and D represent reaction time as a function of presence or absence of LN in the FEV and AEV conditions, respectively.

difficulties with increasing amblyopia severity (controls: 4.73 ± 2.5 ; mild: 3.60 ± 2.2 ; moderate: 2.61 ± 2.03 ; severe: 0.77 ± 1.39 , $P = 0.0001$, ANOVA). We did not find any difference in correctly identified picture differences in amblyopes with or without stereopsis during both FEV (Stereo-present: 4.7 ± 2.2 ; Stereo-absent: 3.5 ± 2.1 , $P = 0.08$ unpaired *t*-test) and AEV (Stereo-present: 2.9 ± 2.2 ; Stereo-absent: 2.9 ± 2.9 , $P = 0.98$ unpaired *t*-test).

We analyzed the reaction time to first correctly identified picture difference during the FEV and AEV. During FEV, reaction time was increased in patients with and without LN (Figs. 2C, 2D; controls: 8243 ± 4055 ; LN absent: $12,247 \pm 6888$; LN present: $14,650 \pm 10,888$, $P = 0.04$, ANOVA). Higher reaction time was seen in patients with increasing amblyopia severity during FEV (Figs. 3C, 3D; controls: 8243 ± 4055 ; mild: $12,401 \pm 8465$; moderate: $14,112 \pm 10,428$; severe: $14,633 \pm 6542$, $P = 0.03$, Kruskal-Wallis ANOVA). A similar trend was seen during AEV. Reaction time was increased in patients with absent stereopsis during FEV (Stereo-present: 9925 ± 5684 msec; Stereo-absent: $17,399 \pm 10,941$ msec, $P = 0.003$, unpaired *t*-test). However, a majority of these patients (6/8) had LN. The reaction time was not increased during AEV (Stereo-present: $10,766 \pm 7490$ msec; Stereo-absent: $11,946 \pm 7447$ msec, $P = 0.68$, unpaired *t*-test). This could be due to fewer data points secondary to fewer correctly identified picture differences.

Both controls and amblyopes had a comparable percentage of fixations in all interest areas during both FEV and AEV, irrespective of amblyopia severity or presence of LN. The total dwell time spent in interest areas containing picture differences were comparable during FEV. However, patients with LN and moderate/severe amblyopia spent less time in interest areas containing differences during AEV. Amongst moderate and severe amblyopes there was a greater correlation between the first interest areas containing a picture difference in which they looked versus first correct click during both FEV and AEV

(Tables 2 and 3). However, the time taken from the first look to click was greater. Fixation duration-based heat maps for all 50 interest areas/visual search trial were constructed. There was no difference in fixation durations in all 50 interest areas/trial among controls and amblyopes with and without LN during both FEV and AEV (Supplementary Table S1). The fixation duration trended down in severe amblyopia during AEV (Supplementary Table S2).

Oculomotor Parameters in Amblyopia Without LN

Frequency of Microsaccades and Saccades. Amblyopes had a greater frequency of microsaccades during FEV compared to AEV (paired *t*-test: 0.0009) during visual fixation, in agreement with previous studies.^{25,39} However, the frequencies of microsaccades (FEV: 0.08 ± 0.05 , AEV: 0.10 ± 0.08 ; paired *t*-test: 0.30) and saccades (FEV: 0.59 ± 0.29 , AEV: 0.44 ± 0.27 ; paired *t*-test: 0.10) were comparable during blank scene viewing, for both FEV and AEV.

The frequencies of microsaccades during the visual search were reduced for both FEV and AEV (controls: 0.39 ± 0.23 Hz; FEV: 0.20 ± 0.14 Hz and AEV: 0.09 ± 0.07 Hz; Kruskal-Wallis ANOVA, $P < 0.0001$). Figures 4A and 4B plot the frequency of microsaccades, whereas 4C and 4D plot the frequencies of saccades during FEV and AEV, respectively. The frequencies of microsaccades declined with increasing amblyopia severity during AEV (controls: 0.39 ± 0.23 Hz; mild: 0.10 ± 0.05 Hz; moderate: 0.13 ± 0.09 Hz; severe: 0.03 ± 0.02 Hz, Kruskal-Wallis ANOVA, $P < 0.0001$). The frequency of saccades was also reduced (controls: 2.9 ± 0.8 Hz; FEV: 2.6 ± 0.4 Hz and AEV: 2.3 ± 0.7 Hz; Kruskal-Wallis ANOVA, $P < 0.0001$). Amblyopes produced fewer saccades during both FEV (controls: 2.9 ± 0.8 Hz; mild: 2.7 ± 0.4 Hz; moderate: 2.7 ± 0.4 Hz; severe: 2.4 ± 0.1 Hz, Kruskal-Wallis ANOVA test, $P = 0.02$) and AEV (controls: 2.8 ± 0.8 Hz; mild: 2.4 ± 0.2 Hz; moderate:

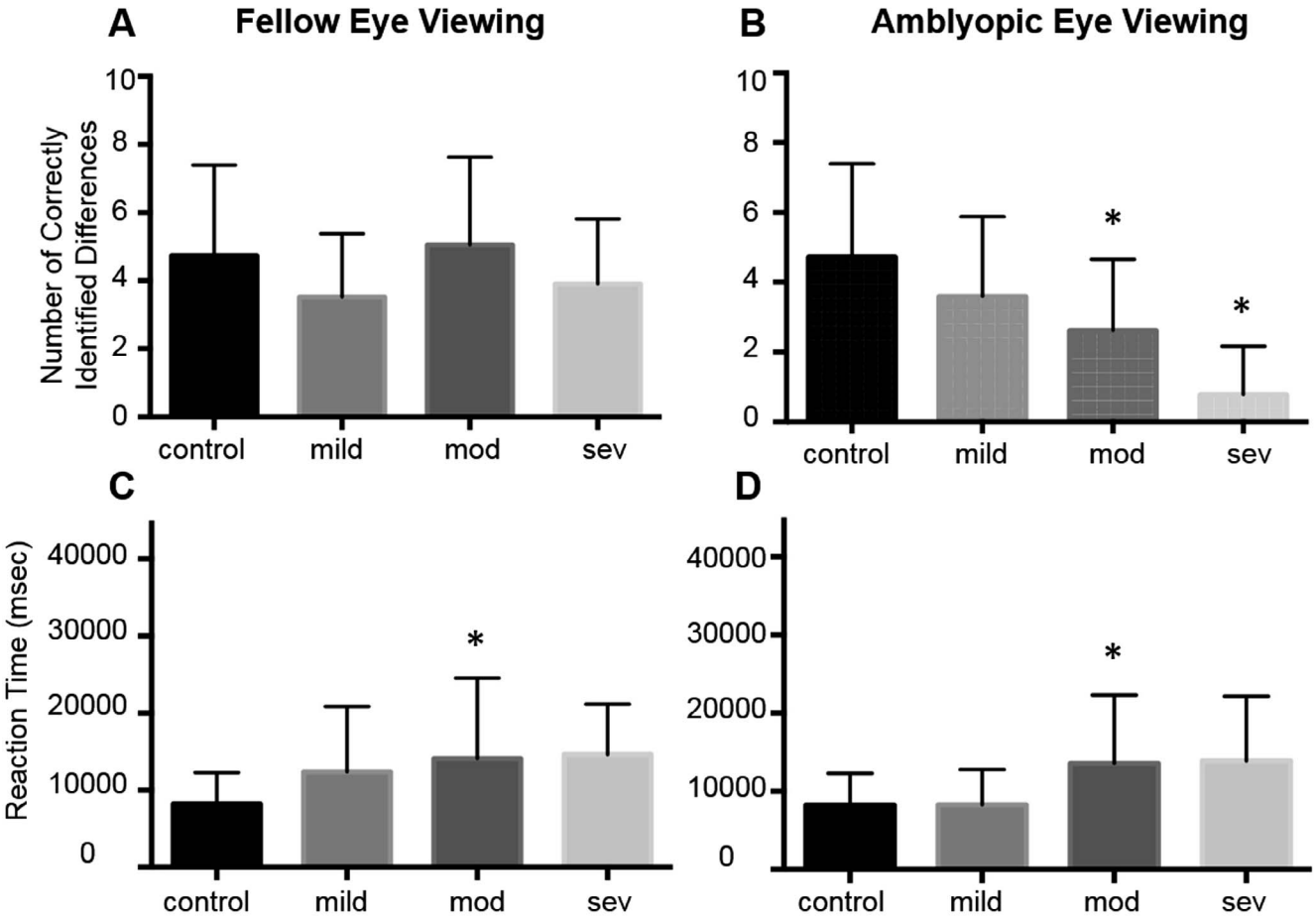


FIGURE 3. A and B represent correctly identified picture differences as a function of severity of amblyopia in the FEV condition and AEV condition, respectively. C and D represent reaction time as a function of severity of amblyopia in the FEV condition and AEV condition, respectively.

2.7 ± 0.8 Hz; severe: 1.5 ± 0.3 Hz, Kruskal-Wallis ANOVA, *P* < 0.0001).

Microsaccade production was greatest during visual fixation followed by visual search and was lowest during blank scene viewing in controls (repeated measures ANOVA: *P* = 0.0082) and FEV of amblyopes (repeated measures ANOVA: *P* = 0.018).

However, during AEV the frequency was greater during visual fixation followed by blank scene viewing, and lowest during visual search (repeated measures ANOVA: 0.035). We found a positive correlation between the frequencies of microsaccades generated during visual fixation versus during visual search (*r* = 0.36, *P* = 0.04, Spearman correlation) in all subjects. This

TABLE 2. Visual Exploration Strategy in Amblyopic Subjects With and Without LN

	Controls	Amblyopia Without LN	Amblyopia With LN	P Value
FEV condition				
First interest area with the picture difference explored corresponds to area of first click (%)	21%	44%	40%	0.11*
Time from first look in any correct interest area to first click (msec)	7640 ± 5355	10,763 ± 7272	14,874 ± 10,737	0.03† ^b
% of fixations in interest areas with picture differences	7.85 ± 3.3	10.3 ± 7.9	8.00 ± 3.5	0.36†
% of trial dwell time spent in correct interest areas	51.94 ± 16.3	49.67 ± 11.71	45.30 ± 20.2	0.39†
AEV condition				
First interest area with the picture difference explored corresponds to area of first click (%)	22%	39%	29%	0.30*
Time from first look in any correct interest area to first click (msec)	8074 ± 5831	10,450 ± 7722	11,326 ± 8270	0.18†
% of fixations in interest area	9.55 ± 7.8	9.7 ± 3.8	8.5 ± 3.49	0.10†
% of trial dwell time spent in correct interest areas	54.26 ± 13.19	44.32 ± 18.44	41.22 ± 15.62	0.0045† ^{a,b}

Dunnett's multiple comparison: a = control versus no LN; b = control versus LN.

* χ^2 test.

† Kruskal Wallis test.

TABLE 3. Visual Exploration Strategy in Amblyopic Subjects as a Function of Severity of Amblyopia

	Controls	Mild	Moderate	Severe	P Value
FEV condition					
First interest area with the picture difference explored corresponds to area of first click (%)	21%	33%	44%	60%	0.09*
Time from first look in any correct interest area to first click	7640 ± 5355	11,722 ± 8571	14,382 ± 12,053	11,816 ± 5066	0.1†
% of fixations in interest area	7.85 ± 3.3	8.03 ± 3.6	11.01 ± 9.1	8.44 ± 3.52	0.36†
% of trial dwell time spent in correct interest areas	51.94 ± 16.3	45.70 ± 18.0	52.45 ± 12.3	42.30 ± 17.4	0.14†
AEV condition					
First interest area with the picture difference explored corresponds to area of first click (%)	22%	16%	52%	33%	0.05*
Time from first look in any correct interest area to first click	8074 ± 5831	8005 ± 3751	13,506 ± 9878	10,766 ± 9001	0.2†
% of fixations in interest area	9.55 ± 7.8	9.19 ± 3.9	8.6 ± 3.0	10.75 ± 4.8	0.18†
% of trial dwell time spent in correct interest areas	54.26 ± 13.9	44.48 ± 15.9	43.02 ± 16.4	38.59 ± 22.37	0.017† ^{b,c}

Dunnett's multiple comparison: a = control versus mild, b = control versus moderate, c = control versus severe, d = mild versus moderate, e = mild versus severe, f = moderate versus severe.

* χ^2 test.

† Kruskal Wallis test.

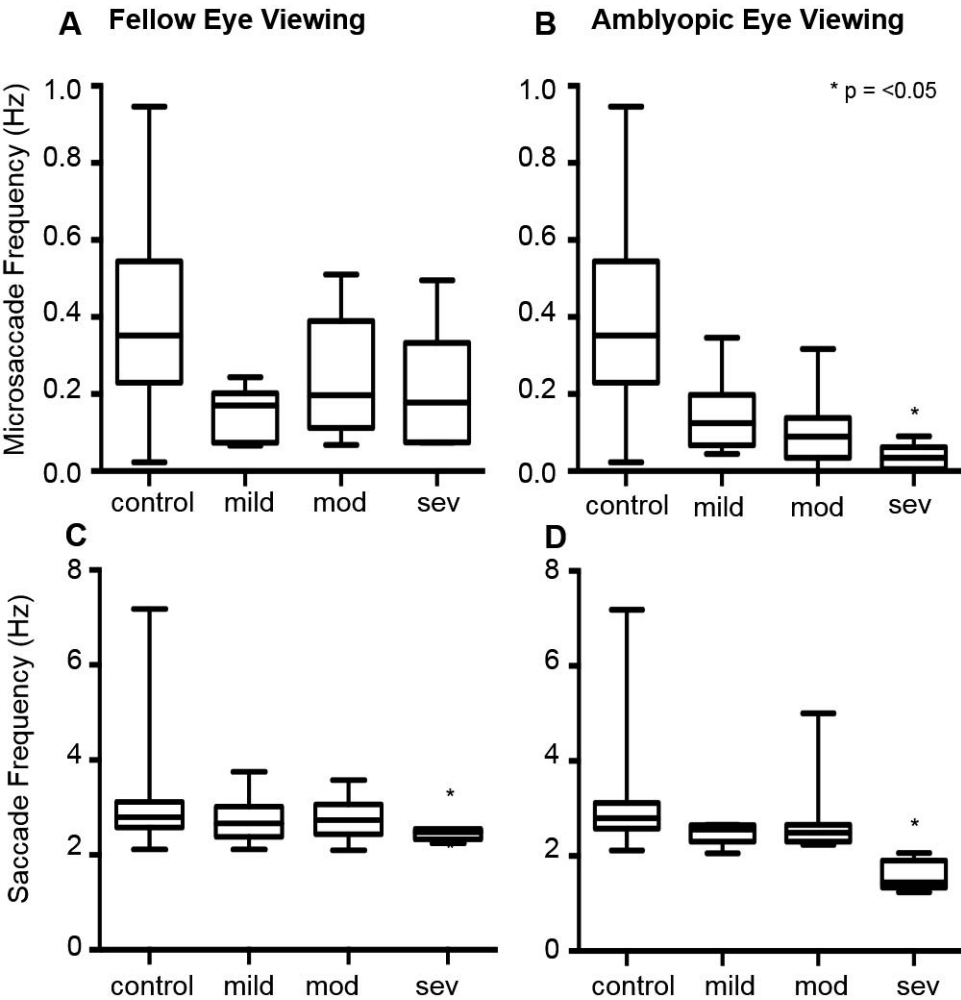


FIGURE 4. Microsaccade frequency during FEV (A) and AEV (B) as a function of severity of amblyopia in subjects without LN during the visual search task. Saccade frequency during FEV (C) and AEV (D) as a function of severity of amblyopia in subjects without LN during the visual search task.

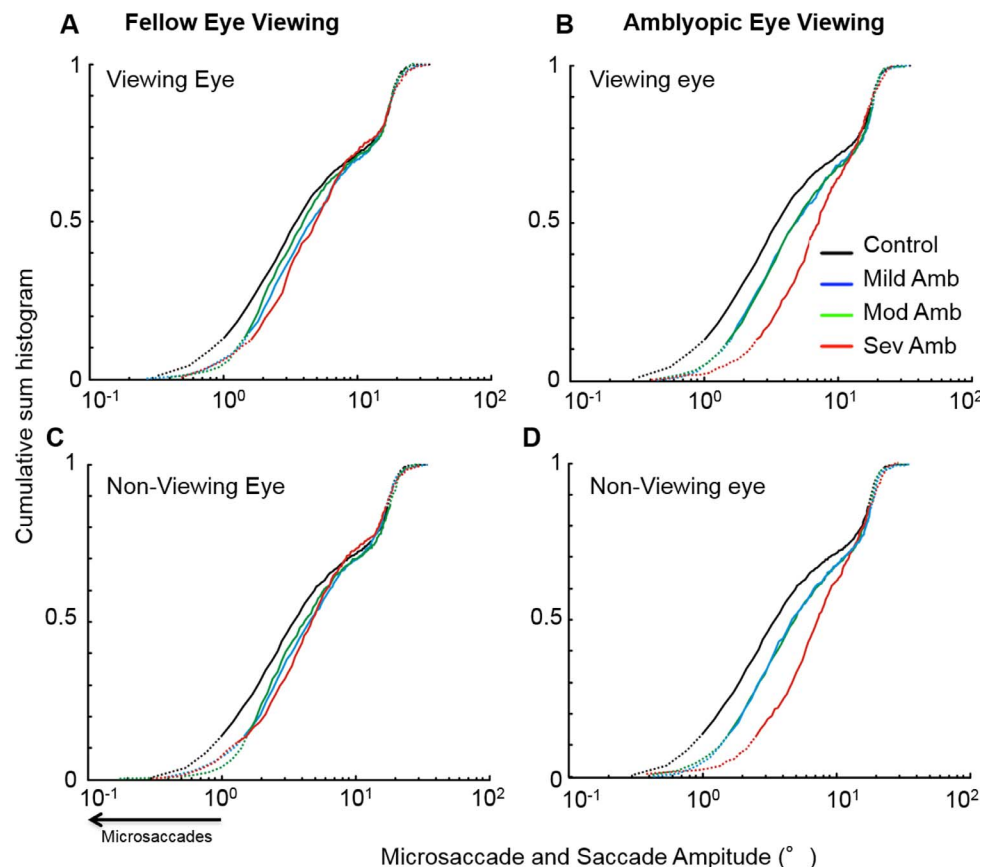


FIGURE 5. **A** and **B** represent cumulative sum histogram of microsaccades and saccades of the viewing eye, whereas **C** and **D** represent nonviewing eye during fellow and amblyopic viewing conditions, respectively, of subjects with amblyopia without LN, produced during the visual search task. Notice the rightward shift of the distribution in both the viewing and nonviewing eye as a function of severity of amblyopia.

correlation lends further support to the hypothesis that similar micro-saccadic behavior is characteristic of fixation and free viewing tasks.⁴

We computed the median amplitude of fixational saccades of the viewing eye and nonviewing eye for each subject during FEV and AEV. We found an increase in the median amplitude of fixational saccade of the viewing eye (FEV: 0.53 ± 0.13 , AEV: 0.90 ± 0.54 ; paired t -test = 0.03) and nonviewing eye (FEV: 0.53 ± 0.13 , AEV: 0.90 ± 0.50 ; paired t -test = 0.02) during AEV, in agreement with previous studies.^{25,39}

We computed the median amplitude of microsaccade/saccades of the viewing and nonviewing eye for each subject during blank scene viewing. Interestingly, we did not find any difference between the median amplitude of microsaccades and saccades of the viewing eye (FEV: 2.8 ± 1.97 , AEV: 1.98 ± 0.78 ; paired t -test = 0.19) and nonviewing eye (FEV: 2.65 ± 1.78 , AEV: 2.21 ± 1.17 ; paired t -test = 0.32) for each subject. Figure 5 summarizes the normalized cumulative sum histogram of the microsaccades and saccades of the viewing eye (Figs. 5A, 5B) and nonviewing eye (Figs. 5C, 5D) elicited during both FEV and AEV. There is a rightward shift of the distribution in both FEV and AEV with increasing amblyopia severity, consistent with decreased frequency of microsaccades and saccades. We computed the percentile of the amplitude of the viewing eye and nonviewing eye for each subject. We then pooled these values for amblyopia by severity. We found that the 10th, 25th, and 50th percentile were different, with larger amplitude in amblyopes during both FEV and AEV (Table 4).

Oculomotor Parameters in Amblyopia With LN

The slow phase velocities were greater in LN patients in both the viewing and nonviewing eye compared to those without LN during all three tasks (Table 5). The slow phase velocities were greater in viewing eye (FEV: $1.33 \pm 2.10^\circ/\text{s}$, AEV: $3.03 \pm 6.17^\circ/\text{s}$; Wilcoxon test: $P = 0.0059$) and nonviewing eye (FEV: $1.80 \pm 2.78^\circ/\text{s}$, AEV: $3.83 \pm 8.00^\circ/\text{s}$; Wilcoxon test: $P = 0.0390$) in AEV, during visual fixation. The slow phase velocities were comparable in the viewing eye (FEV: $2.0 \pm 1.7^\circ/\text{s}$, AEV: $1.2 \pm 0.42^\circ/\text{s}$; Wilcoxon test: $P = 0.21$) and nonviewing eye (FEV: $1.82 \pm 1.03^\circ/\text{s}$, AEV: $1.64 \pm 0.95^\circ/\text{s}$; Wilcoxon test: $p = 0.81$) during blank scene viewing.

Slow phase velocities were greater in the viewing eye during AEV ($2.95 \pm 4.21^\circ/\text{s}$) compared to FEV ($1.97 \pm 2.77^\circ/\text{s}$, Wilcoxon test: $P = 0.0006$) during visual search. However, the slow phase velocities of the nonviewing eye were comparable during FEV and AEV (FEV: $3.05 \pm 3.57^\circ/\text{s}$, AEV: $3.37 \pm 4.38^\circ/\text{s}$; Wilcoxon test: $P = 0.31$). In amblyopes without LN, the intersaccadic drift of the viewing eye (mild: $0.92 \pm 0.63^\circ/\text{s}$; moderate: $1.07 \pm 0.81^\circ/\text{s}$; severe: $1.13 \pm 1.04^\circ/\text{s}$, ANOVA $P < 0.0001$) and nonviewing eye (mild: $1.04 \pm 0.78^\circ/\text{s}$; moderate: $1.3 \pm 1.06^\circ/\text{s}$; severe: $1.15 \pm 1.09^\circ/\text{s}$, ANOVA $P < 0.0001$) during FEV increased with amblyopia severity. Similarly, the intersaccadic drift of the viewing eye (mild: $0.94 \pm 0.68^\circ/\text{s}$; moderate: $1.10 \pm 0.93^\circ/\text{s}$; severe: $1.12 \pm 1.01^\circ/\text{s}$, 1-way ANOVA $P = 0.04$) and nonviewing eye (mild: $1.05 \pm 0.87^\circ/\text{s}$; moderate: $1.3 \pm 1.3^\circ/\text{s}$; severe: $1.4 \pm 1.2^\circ/\text{s}$, 1-way ANOVA $P < 0.0001$) during AEV increased with amblyopia severity. There

TABLE 4. Percentile Amplitude of Microsaccades and Saccades in Viewing Eye and Nonviewing Eye of Amblyopic Subjects Without LN as a Function of Severity of Amblyopia

Percentiles	Controls	Mild	Moderate	Severe	Kruskal Wallis ANOVA <i>P</i> Value
FEV condition					
10th percentile	1.04° ± 0.29° (1.05 ± 0.26)	1.45 ± 0.37 (1.44 ± 0.45)	1.25 ± 0.30 (1.33 ± 0.27)	1.64 ± 0.86 (1.54 ± 0.91)	0.0035 ^{a,c} (0.0010) ^{a,c}
25th percentile	1.81 ± 0.47 (1.76 ± 0.46)	2.28 ± 0.39 (2.30 ± 0.43)	2.01 ± 0.30 (2.03 ± 0.31)	2.66 ± 0.72 (2.66 ± 0.78)	0.0015 ^{a,c} (0.0006) ^{a,c}
50th percentile	3.72 ± 0.91 (3.65 ± 0.91)	4.82 ± 1.10 (4.81 ± 1.18)	4.08 ± 0.92 (4.07 ± 0.90)	5.10 ± 1.62 (4.96 ± 1.40)	0.02 ^{a,c} (0.01) ^{a,c}
75th percentile	12.54 ± 3.25 (12.50 ± 3.22)	13.55 ± 2.44 (13.10 ± 2.44)	13.48 ± 4.10 (13.15 ± 3.75)	11.68 ± 3.15 (11.20 ± 3.07)	0.60 (0.52)
90th percentile	18.22 ± 1.63 (18.15 ± 1.57)	18.41 ± 2.03 (18.06 ± 1.91)	19.19 ± 0.75 (18.73 ± 1.06)	18.14 ± 3.05 (17.44 ± 2.77)	0.07 (0.42)
AEV condition					
10th percentile	1.05 ± 0.29 (1.10 ± 0.26)	1.58 ± 0.30 (1.49 ± 0.32)	1.58 ± 0.58 (1.57 ± 0.60)	2.56 ± 0.92 (2.55 ± 0.79)	<0.0001 ^{a,b,c} (<0.0001) ^{a,b,c}
25th percentile	1.84 ± 0.45 (1.81 ± 0.47)	2.40 ± 0.55 (2.47 ± 0.46)	2.48 ± 0.68 (2.56 ± 0.69)	4.35 ± 1.20 (4.32 ± 1.24)	<0.0001 ^{a,b,c} (<0.0001) ^{a,b,c}
50th percentile	3.82 ± 0.95 (3.72 ± 0.93)	4.98 ± 1.53 (4.81 ± 1.55)	5.08 ± 1.22 (5.29 ± 1.24)	7.86 ± 1.68 (7.90 ± 1.86)	<0.0001 ^{a,b,c} (<0.0001)
75th percentile	12.75 ± 3.45 (12.61 ± 3.49)	13.80 ± 3.18 (13.94 ± 3.26)	14.72 ± 2.25 (15.17 ± 2.72)	14.06 ± 2.52 (13.96 ± 2.66)	0.20 (0.12)
90th percentile	18.54 ± 1.56 (18.35 ± 1.54)	17.83 ± 3.80 (17.99 ± 4.24)	18.36 ± 1.04 (18.66 ± 1.92)	19.29 ± 2.03 (19.46 ± 2.12)	0.37 (0.27)

All the values in parenthesis are amplitude of the nonviewing eye. Dunnett's multiple comparison: a = controls versus mild, b = control versus moderate, c = control versus severe, d = mild versus moderate, e = mild versus severe, f = moderate versus severe.

was no such systematic increase in the slow phase velocities of patients with LN with amblyopia severity (viewing eye [mild: $1.3 \pm 1.6^\circ/\text{s}$; moderate: $4.8 \pm 6.6^\circ/\text{s}$; severe: $1.2 \pm 0.71^\circ/\text{s}$] and nonviewing eye [mild: $2.08 \pm 2.2^\circ/\text{s}$; moderate: $7.8 \pm 8.8^\circ/\text{s}$; severe: $2.1 \pm 1.8^\circ/\text{s}$]) during FEV (viewing eye [mild: $1.6 \pm 1.3^\circ/\text{s}$; moderate: $5.4 \pm 7.7^\circ/\text{s}$; severe: $2.08 \pm 1.2^\circ/\text{s}$] and nonviewing eye [mild: $2.03 \pm 1.86^\circ/\text{s}$; moderate: $6.0 \pm 7.3^\circ/\text{s}$; severe: $2.9 \pm 1.7^\circ/\text{s}$]) and during AEV.

Intersaccadic drift was greatest during visual search followed by blank scene viewing and least during visual fixation in both the viewing eye (repeated measures ANOVA: $P = 0.01$) and nonviewing eye (repeated measures ANOVA: $P = 0.01$) of controls. A similar result was seen in both the viewing eye (FEV: repeated measures ANOVA $P = 0.001$; AEV: repeated measures ANOVA $P = 0.04$) and nonviewing eye (FEV: repeated measures ANOVA $P = 0.02$; AEV: repeated measures ANOVA $P = 0.007$) in amblyopes without LN. The slow phase velocity was comparable in the viewing eye during both FEV and AEV for all three tasks (FEV: repeated measures ANOVA $P = 0.1$; AEV: repeated measures ANOVA $P = 0.1$). The slow phase velocity was greatest in the nonviewing eye during visual search followed by blank scene viewing and least during visual fixation (FEV: repeated measures ANOVA $P = 0.04$; AEV: repeated measures ANOVA $P = 0.006$).

Amblyopes had a comparable frequency of quick phases during FEV and AEV during visual fixation (FEV: 1.3 ± 0.9 Hz; AEV: 1.14 ± 0.6 Hz paired t -test = 0.39) and blank scene viewing (FEV: 1.3 ± 0.5 Hz; AEV: 1.09 ± 0.5 Hz paired t -test = 0.25). We assessed the frequency of quick phases of LN and saccadic events during visual search. The frequencies were comparable during FEV (mild: 2.5 ± 0.71 Hz; moderate: 3.20 ± 0.59 Hz; severe: 1.9 ± 1.3 Hz; 1-way ANOVA $P = 0.09$) and AEV (mild: 2.7 ± 0.68 Hz; moderate: 2.6 ± 0.79 Hz; severe: 2.08 ± 0.04 Hz; 1-way ANOVA: $P = 0.25$).

We computed the median amplitude of quick phases and saccades of the viewing eye and nonviewing eye during blank scene viewing. We did not find any difference between the median amplitude of the viewing eye (FEV: $2.7^\circ \pm 1.3^\circ$ and AEV: $3.3^\circ \pm 2.2^\circ$; paired t -test = 0.51) and nonviewing eye (FEV: $3.3^\circ \pm 2.2^\circ$ and AEV: $2.7^\circ \pm 1.3^\circ$; paired t -test = 0.47) for each subject. Figure 4 summarizes the normalized cumulative sum histogram of the quick phases and saccades of the viewing eye (Figs. 6A, 6B) and nonviewing eye (Figs. 6C, 6D) elicited during FEV and AEV as a function of amblyopia severity. There is a rightward shift of the distribution of amplitude of the viewing eye and nonviewing eye during AEV. We found that the 10th, 25th, and 50th percentile were different, with larger amplitude in amblyopes during AEV (Table 6).

DISCUSSION

The main findings of our study are (1) amblyopes had greater difficulties performing the visual search, during both fellow and AEV, compared to controls. They identified fewer picture differences, took longer to identify differences, and task performance was worse with increased severity of amblyopia. Additionally, amblyopes spent less time looking at IAs containing picture differences. (2) Amblyopes without LN had significantly decreased microsaccade (the physiological fixational saccade) and saccadic frequency during fixation and free-viewing scene tasks. (3) LN exacerbated the deficits noted among amblyopes.

Prior research in healthy subjects has found that microsaccades are generated with greater frequency with increasing complexity of visual scenes and nearer the identified target.^{4,35} Little is known about visual search in amblyopia. The visual cortex signals microsaccade production while the superior

TABLE 5. Slow Phase Velocity in Amblyopes With LN and Intersaccadic Drifts in Amblyopes Without LN During Visual Fixation, Viewing a Blank Scene and While Performing Visual Search Tasks

	Control	Fellow Eye No LN	Amblyopic Eye No LN	Fellow Eye LN	Amblyopic Eye LN	P Value
Viewing eye (°/s)	0.62 ± 0.79 (0.89 ± 0.73) [1.09 ± 0.81]	0.67 ± 0.67 (0.70 ± 0.54) [1.03 ± 0.80]	0.65 ± 0.55 (0.89 ± 1.00) [1.08 ± 0.91]	2.7 ± 3.7 (2.85 ± 4.83) [2.78 ± 4.81]	5.02 ± 8.11 (2.27 ± 2.44) [3.08 ± 5.157]	<0.0001 ^{c,d} (< 0.0001) ^{c,d} [< 0.0001] ^{c,d}
Nonviewing eye (°/s)	0.53 ± 0.67 (0.99 ± 0.75) [1.3 ± 1.2]	0.75 ± 0.68 (0.75 ± 0.66) [1.18 ± 0.98]	0.68 ± 0.52 (0.91 ± 0.78) [1.26 ± 1.20]	3.4 ± 4.6 (3.6 ± 4.8) [4.4 ± 6.40]	5.9 ± 10.16 (2.0 ± 1.56) [3.58 ± 5.10]	<0.0001 ^{c,d} (< 0.0001) ^{c,d} [< 0.0001] ^{c,d}

Round parenthesis () are values while viewing a blank scene and box parenthesis [] are values while performing a visual search task. Dunnett's multiple comparison: a = control versus fellow no LN, b = control versus amb no LN, c = control versus fellow LN, d = control versus amb LN.

colliculus drives such input to generate microsaccades.^{40–42} Therefore, visual cortical impairment likely affects the physiology of subcortically generated microsaccades. Studies have shown that during fixation, amblyopes have decreased microsaccade production.^{25,39} Microsaccade are thought to provide an optimal sampling method for the brain to improve high-frequency spatial resolution during free-scene viewing.^{43–45} We found reduced microsaccade and saccade frequencies during visual search, which correlated with increasing amblyopia severity. Decreased frequency of microsaccades could diminish the ability to perceive fine details and lead to impaired perception of spatially adjacent small objects, a phenomenon known as “crowding.” Similar to controls, microsaccades and saccades followed the main sequence in amblyopia. These findings support the theory that both microsaccades and

exploratory saccades have a common neurophysiologic basis.^{4,46,47}

Previous studies have reported abnormalities in the fellow and amblyopic eye including reduced contrast,^{13,14,48,49} spatial integration,^{50–53} global motion,^{54–57} and motion-perception.^{58–60} These studies did not categorize patients with and without LN. We found that amblyopes, particularly those whom develop the disease earlier in infancy as demonstrated by the presence of LN,^{24,61–65} had greater difficulty performing a visual search. We found increased intersaccadic drifts in LN patients compared to those without LN. This is likely due to the contribution of the pathologic slow phase velocity of LN. It is known that all types of manifest LN are strongly visually driven. Removal of the visual target is associated with decreased slow phase velocity and reduced frequency of quick

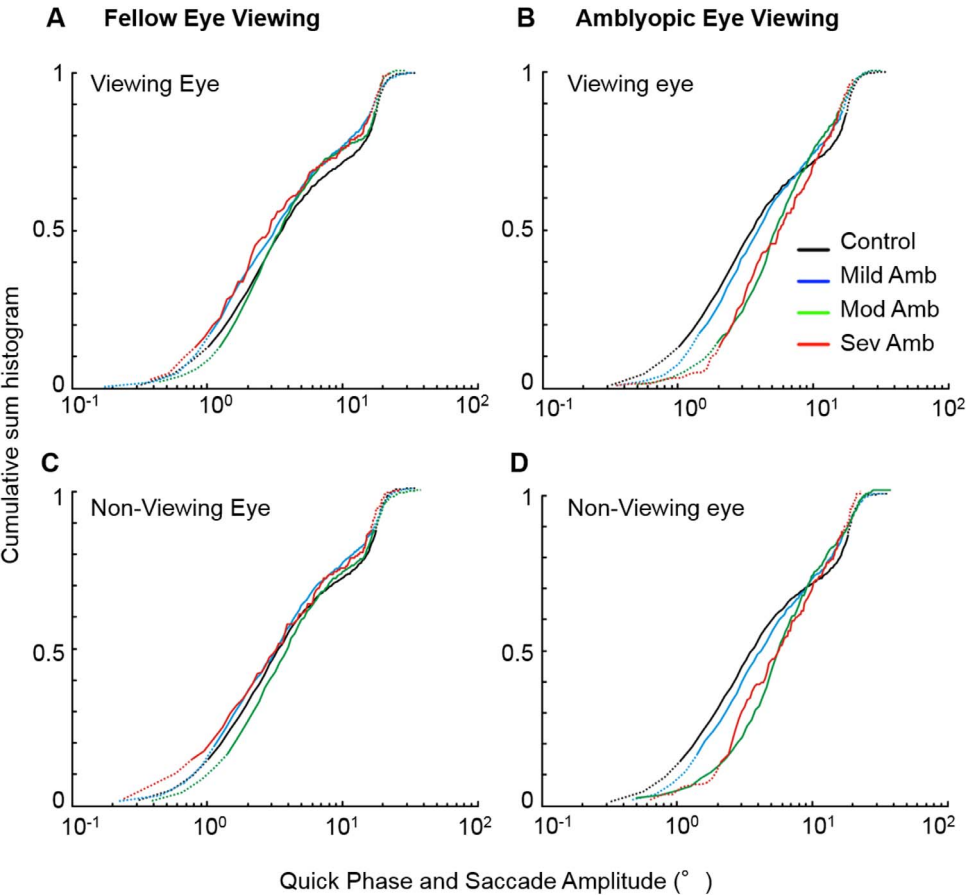


FIGURE 6. A and B represent cumulative sum histogram of quick phases of LN and saccades of the viewing eye, whereas C and D represent nonviewing eye during fellow and amblyopic viewing conditions respectively, of subjects with amblyopia with LN, produced during the visual search task.

TABLE 6. Percentile Amplitude of Quick Phases and Saccades in Viewing Eye and Nonviewing Eye of Amblyopic Subjects With LN as a Function of Severity of Amblyopia

Percentiles	Controls	Mild	Moderate	Severe	Kruskal Wallis ANOVA P Value
FEV condition					
10th percentile	1.04 ± 0.29 (1.05 ± 0.26)	0.98 ± 0.20 (0.96 ± 0.17)	1.21 ± 0.38 (1.23 ± 0.44)	0.77 ± 0.12 (0.56 ± 0.21)	0.17 (0.11)
25th percentile	1.81 ± 0.47 (1.76 ± 0.46)	1.65 ± 0.52 (1.65 ± 0.32)	1.88 ± 0.38 (1.98 ± 0.54)	1.27 ± 0.21 (1.29 ± 0.20)	0.15 (0.32)
50th percentile	3.72 ± 0.91 (3.65 ± 0.91)	3.60 ± 1.17 (3.52 ± 0.78)	3.61 ± 0.57 (3.96 ± 0.56)	2.87 ± 1.06 (3.05 ± 1.04)	0.75 (0.38)
75th percentile	12.54 ± 3.25 (12.50 ± 3.22)	9.10 ± 3.26 (9.32 ± 3.06)	12.45 ± 4.87 (12.80 ± 4.06)	8.88 ± 7.26 (9.73 ± 5.97)	0.06 (0.06)
90th percentile	18.22 ± 1.63 (18.15 ± 1.57)	17.56 ± 1.59 (17.90 ± 1.69)	15.95 ± 4.16 (16.84 ± 3.59)	12.52 ± 8.14 (12.45 ± 6.8)	0.39 (0.25)
AEV condition					
10th percentile	1.05 ± 0.29 (1.10 ± 0.26)	1.36 ± 0.43 (1.26 ± 0.51)	2.43 ± 1.07 (2.52 ± 1.08)	2.08 ± 0.58 (1.99 ± 0.46)	0.0004 ^{b,c} (0.0006) ^{b,c}
25th percentile	1.84 ± 0.45 (1.81 ± 0.47)	2.10 ± 0.55 (2.05 ± 0.60)	3.77 ± 1.47 (3.70 ± 1.47)	2.87 ± 0.74 (2.79 ± 0.42)	0.0023 ^b (0.0024) ^b
50th percentile	3.82 ± 0.95 (3.72 ± 0.93)	4.49 ± 1.38 (4.45 ± 1.40)	5.72 ± 2.21 (5.59 ± 1.97)	5.54 ± 2.06 (5.60 ± 1.60)	0.03 ^b (0.017) ^b
75th percentile	12.75 ± 3.45 (12.61 ± 3.49)	10.66 ± 2.79 (10.84 ± 3.26)	10.84 ± 2.46 (10.71 ± 2.81)	11.12 ± 1.06 (11.32 ± 0.79)	0.37 (0.41)
90th percentile	18.54 ± 1.56 (18.35 ± 1.54)	17.26 ± 1.65 (17.55 ± 2.02)	16.63 ± 2.70 (16.50 ± 3.06)	16.80 ± 1.78 (17.50 ± 0.02)	0.16 (0.64)

All the values in parenthesis are amplitude of the nonviewing eye. Dunnett's multiple comparison: a = control versus mild, b = control versus moderate, c = control versus severe.

phases.⁶⁶ We did not find an increase in slow phase velocity with amblyopia severity. Thus, it is possible that in severe amblyopia, poor vision could have resulted in decreased slow phase velocity. Since the quick phases of LN were interspersed with saccades, we did not find reduced frequency of saccades in LN subjects. Our recordings were performed under monocular viewing, which enabled us to accurately assess the presence of LN. The prevalence of manifest LN (present in monocular and binocular viewing) is 10% to 15%, whereas true LN (present only under monocular viewing) is rare. Thus, it is likely that majority of our patients had manifest LN. Thus, while it is possible that the deficits described here were exacerbated due to an increase in nystagmus under monocular viewing, we would expect to see similar deficits in these patients during binocular viewing.

Our results are in agreement with other studies evaluating reading^{67,68} and reaching in amblyopia.⁶⁹ Amblyopes have reading difficulties, with slower reading related to fellow eye fixation instability. Amblyopes with abnormal binocularity showed impairments in critical aspects of motor control such as speed and accuracy during real-world visuomotor tasks, the extent of which correlated with the loss of stereoacuity.⁷⁰ We found a similar correlation of increased difficulties in visual search in patients with LN and severe amblyopia, both of these groups are known to have increased fixation instability^{20,25,71} and poor stereopsis.⁷²

Multiple studies have shown that amblyopes exhibit higher order perceptual deficits that suggest abnormal processing in both the ventral “what” and the dorsal “action” pathways and deficits in respect to completion of tasks involving higher order attentional components.^{73–76} We did not examine how these deficits impact the ability to perform a visual search. As microsaccades are known to increase with attentional load,^{77–79} poorer attention might explain why amblyopes were found to have decreased microsaccade frequencies during the search. Future research should be directed at determining the influence of attention and other visual functions on the kinematic deficits seen during visual search in amblyopia.

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